

Elbow Injuries in Pediatric and Adolescent Athletes

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RELEVANT ANATOMY AND BIOMECHANICS

Osteology

Skeletal growth around the elbow typically follows a characteristic development process. The elbow joint consists of the articulation of the distal humerus with the ulna (the ulnohumeral joint), the articulation of the distal humerus with the radial head (the radiocapitellar joint), and the proximal articulation of the radius and ulna (the proximal radioulnar joint). Skeletal maturation occurs from the primary ossification centers of the humerus, radius, and ulna, along with six secondary ossification centers. The sequence of ossification is the capitellum, proximal radius, medial epicondyle, trochlea, olecranon, and lateral epicondyle.¹⁻³ When evaluating radiographs, variation can occur in the appearance of the ossification centers as distinct structures, and correlation with the contralateral extremity can help determine disease in questionable cases.

The ossification of the distal humerus extends to the condyles at birth and progresses through the secondary centers at various stages of development. The lateral condyle and capitellum appear in the second year of life.⁴ On a normal lateral radiograph of the elbow, the anterior humeral line intersects the anterior third of the capitellar ossific nucleus (Fig. 134.1). This line is helpful in characterizing supracondylar fracture displacement in children. In the third year of life, the proximal radius ossific nucleus begins to ossify and is typically present in most children by age 4 years. Notches or clefts in the proximal radius metaphysis can sometimes be seen and are considered part of the normal variability of maturation.^{5,6} The medial epicondylar ossific nucleus begins to ossify between the ages of 5 and 6 years; however, fusion of the epiphysis does not occur until 15 to 16 years of age, placing this physis under stress during the throwing motion well into adolescence.⁷ The secondary ossification center of the olecranon process appears around age 7 to 9 years, whereas the trochlear ossific nucleus appears around age 9 to 10 years.⁸ The last center to appear is the lateral epicondyle, usually after age 10 years, and this center rapidly fuses to the lateral condyle shortly after being visible on radiographs. A thorough understanding of the stages of ossification and the appearance of the centers of ossification is critical for clinicians who treat pediatric elbow injuries (Fig. 134.2). Any heterogeneous appearance to the nuclei or differences relative to the contralateral extremity in terms of size, density, position, or fragmentation could indicate abnormal development

and may be the result of repetitive stress to the elbow, inducing vascular changes with resultant alterations in the maturation process. A failure to appreciate the normal progression of growth at the elbow could result in misdiagnosis, improper treatment, and progressive developmental abnormality.

The osseous anatomy surrounding the elbow contributes significantly to stability of the elbow joint. Approximately 50% of elbow stability arises from the congruity of the ulnohumeral articulation.⁹ In serial olecranon excision studies, progressive loss of the proximal olecranon resulted in linear decreases in elbow stability at 0 and 90 degrees of elbow motion.¹⁰ The radial head provides 15% to 30% of valgus stability to the elbow joint, and may be more important in the throwing athlete for providing valgus stability through the midrange of motion.⁹ Studies evaluating the load transmission across the elbow joint have shown that forces of up to three times body weight can be seen, and forces that are generated during throwing may be even higher.^{11,12} Thus it is not surprising that a small deficiency in the elaborate stability-controlling mechanisms of the elbow may have a significant and cumulative effect on elbow function.

Ligaments and Soft Tissue

Whereas 50% of the stability of the elbow joint arises from the osseous anatomy, the remaining 50% is derived from the ligamentous and soft tissue attachments (e.g., the anterior capsule, the medial or ulnar collateral ligament [UCL], and the lateral radial collateral ligament).⁹ With regard to the muscular attachment sites around the elbow, laterally the common extensor origin is located at the lateral epicondyle and can be a common source of pathologic findings in persons who participate in racquet sport and repetitive motion activities. This source is perhaps more prominent in the adult population, with abnormalities classically located within the extensor carpi radialis brevis (lateral epicondylitis). Anteriorly, the brachialis and biceps serve as a powerful flexor and supinator of the elbow, respectively. Posteriorly, the triceps attachment on the proximal aspect of the olecranon serves as the main extensor to the elbow and can be injured in young athletes who have an eccentric contractile injury. Although injury or strain of these muscle groups can occur, this pathologic condition is less often seen in the pediatric population. Medially, the flexor and pronator attachment sites serve as powerful medial stabilizers to the elbow, particularly in the athlete

Abstract

Elbow injuries in athletic children and adolescents are a distinct clinical entity when compared with the adult population. The presence of the distal humeral physis, multiple ossification centers, developing muscular and ligamentous attachments, and articulating cartilage can contribute to characteristic pathologic conditions in vulnerable patient populations. Based on the age of the patient, the duration and location of symptoms, and the specific sport, the clinician can arrive at the diagnosis and develop a clear treatment strategy. Nonoperative treatment often can result in a high degree of success when treating this patient population; however, several conditions may require surgical intervention.

Keywords

elbow
adolescent
pediatric
athlete
injury

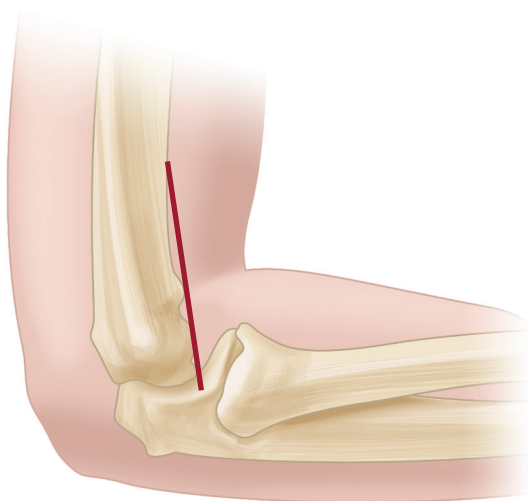


Fig. 134.1 On a normal lateral radiograph of the pediatric elbow, the anterior humeral line intersects the anterior third of the capitellum and is helpful in characterizing displacement in supracondylar humerus fractures.

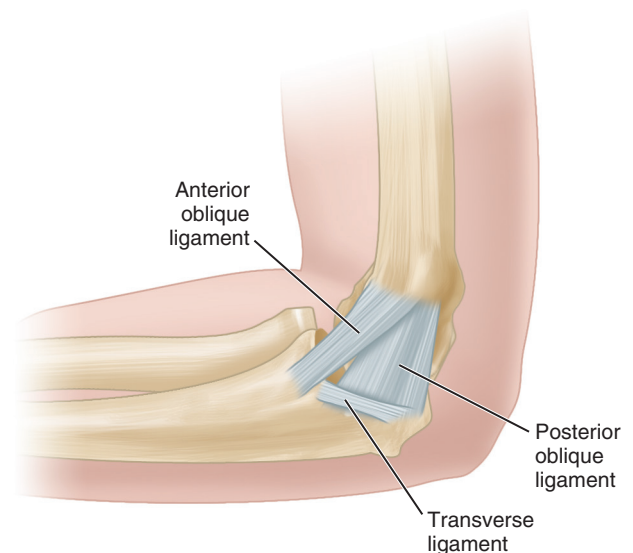


Fig. 134.3 The ulnar collateral ligament of the elbow consists of the anterior oblique bundle, the intermediate bundle, and the posterior oblique bundle.

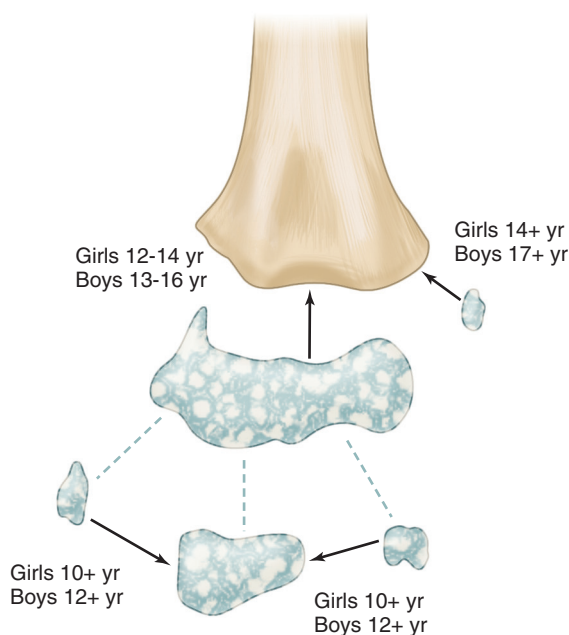


Fig. 134.2 The usual ages at which the ossific centers fuse to each other and to the distal humerus in boys and girls.

whose sport entails throwing. Repetitive strain at this site can be seen in the throwing population or, alternatively, in golfers, in whom it can manifest as medial epicondylitis.

The UCL complex of the elbow is a broad ligament on the medial side of the elbow that provides restraint to valgus stress (Fig. 134.3).¹³ The ligament consists of three portions: the anterior oblique bundle, the posterior oblique bundle, and the transverse ligament or intermediate bundle. The anterior oblique bundle is the main medial stabilizer to the elbow and arises from the medial epicondyle with an insertion to the medial aspect of the

coronoid process. This ligament becomes more important during high-velocity throwing, when a large valgus force is placed on the medial elbow structures. Both the anterior and posterior bundles of the ligament provide stability during the full arc of motion, with the anterior band having a larger burden in extension and the posterior band having a more important role in flexion. The lateral collateral ligament provides varus stability to the elbow, and in recent years its role as a major contributor to posterolateral rotatory elbow stability has been more extensively studied.¹⁴ The lateral collateral (or radial collateral) ligament complex is composed of three main parts: the lateral UCL, the radial collateral ligament, and the accessory lateral collateral ligament (Fig. 134.4). The radial collateral ligament courses from the lateral epicondyle to the annular ligament of the proximal radioulnar joint. The lateral UCL courses from the posterior lateral epicondyle, travels across the annular ligament, and attaches on the crista supinatoris of the ulna. This ligament is the primary restraint to posterolateral elbow stability. The accessory lateral collateral ligament originates from the inferior aspect of the annular ligament and attaches to the tubercle of the supinator. The lateral ligamentous complex is classically injured or disrupted during an elbow dislocation or iatrogenically during surgery around the posterolateral aspect of the elbow. These ligaments, combined with the anconeus muscle, form a lateral elbow complex that provides both dynamic and static restraint to elbow instability in both varus and rotation. The lateral ligament complex is rarely injured in children as a result of repetitive use or microtrauma; however, the medical literature includes case reports that describe reconstruction or repair of this complex when persistent instability exists.^{15,16} The diagnosis is often challenging because children may describe vague pain and soreness with certain activities, but without the characteristic instability that is the hallmark of this clinical entity in adults.

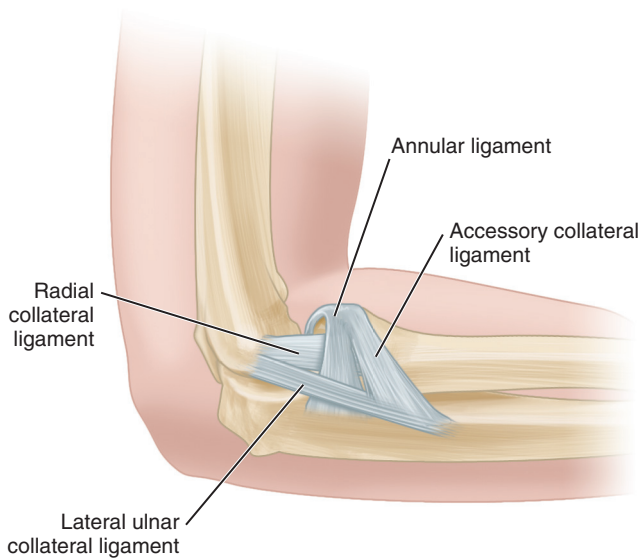


Fig. 134.4 The lateral collateral ligament complex of the elbow is demonstrated, including the lateral ulnar collateral ligament, the radial collateral ligament, and the accessory collateral ligament.



Fig. 134.6 The early cocking phase of throwing.

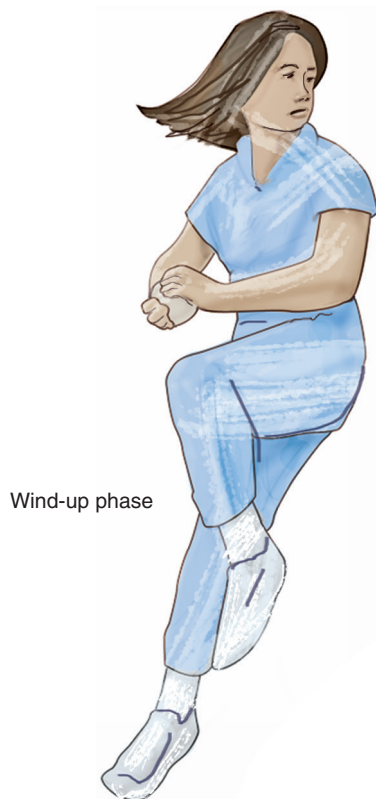


Fig. 134.5 The wind-up phase of throwing.

Biomechanics in Throwing Sports

Throwing Motion

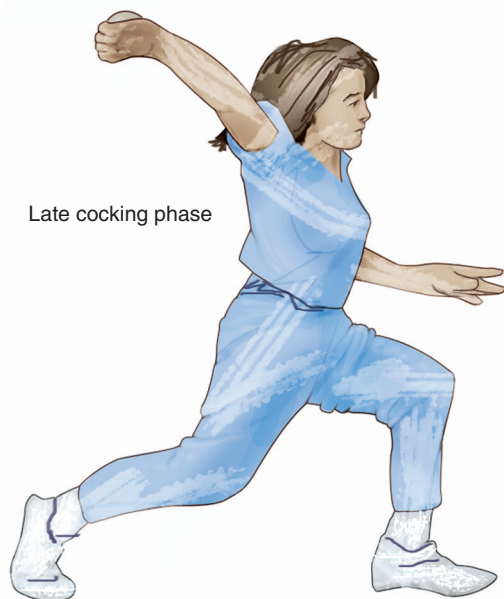
The throwing motion is common in many sports played by children and adolescents. Football, baseball, and overhead racquet sports such as tennis are the characteristic activities that demonstrate the biomechanics of the elbow during throwing and

are most common in this age group. The pitching motion has been analyzed extensively in the adult and pediatric sports medicine literature. Repetitive valgus and distraction forces of the medial pediatric elbow have been observed during pitching and are thought to be the etiology in the development of medial UCL disease, medial epicondylitis, and epicondyle avulsions. The pitch has been divided into five stages^{17–19}:

1. Wind-up (ends when the ball leaves the nonthrowing hand; Fig. 134.5)
2. Cocking
 - a. Early cocking (shoulder abduction and rotation; ends when the forward foot hits the ground; Fig. 134.6)
 - b. Late cocking (maximum shoulder external rotation is achieved; Fig. 134.7)
3. Acceleration (begins with internal rotation of the humerus and ends with ball release; Fig. 134.8)
4. Deceleration (eccentric contraction of all muscles; center of gravity moves over plantar foot)
5. Follow-through (ends when all motion is complete; Fig. 134.9)

One of the fundamental concepts in understanding the throwing motion as it relates to pitching is the “kinetic chain.”²⁰ The kinetic chain of throwing refers to the role that core and proximal musculature play into the development of force required for throwing. Proximal muscle activation results in postural adjustments that allow the body to balance the forces that are required for throwing.²¹ These muscle activations create force moments, which are a product of adjacent segment motion and position. The moments occur at proximal segments and are important for generating force delivered to distal motion segments, such as the elbow.²² One investigation demonstrated that younger pitchers may not produce as much force generation from their lower core segments, which may place larger demands on the upper extremities of youths who pitch and can ultimately cause

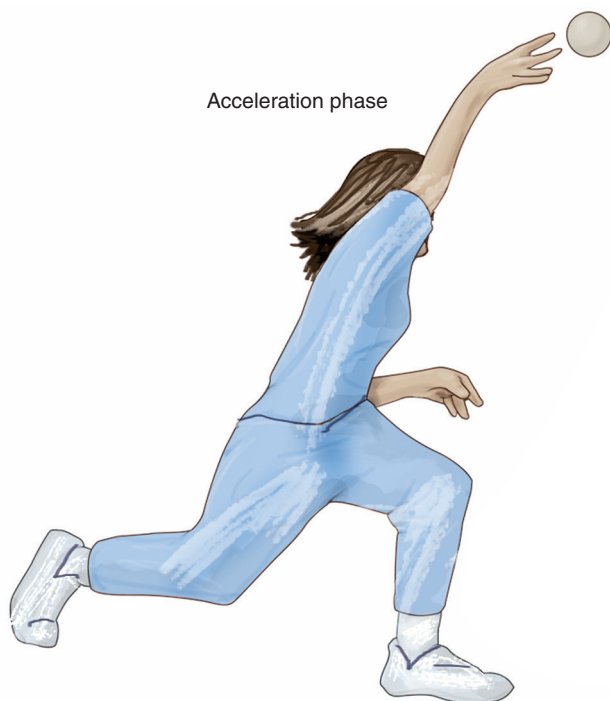
Late cocking phase

**Fig. 134.7** The late cocking phase of throwing.

Follow-through phase

**Fig. 134.9** The follow-through phase of throwing.

Acceleration phase

**Fig. 134.8** The acceleration phase of throwing.

pain and disability.²³ Other investigations have documented that youths who pitch and adults who pitch have similar kinematics, while also demonstrating that youth pitchers who are deficient in their mechanics on key parameters are less efficient, place higher valgus loads across their elbows, and demonstrate higher humeral internal rotation torque.^{24,25} Lack of proximal force production may cause the elbow to be lower than the shoulder during the acceleration phase of throwing (as a result of a lack

of elbow elevation and extension prior to shoulder rotation), which places increased tensile forces on the medial elbow.²⁰ In the ideal motion, maximal elbow extension occurs prior to maximum shoulder rotation, which couples internal rotation of the shoulder to pronation about the elbow, preventing medial tensile or valgus overload. Teaching proper throwing mechanics in youth sports and limiting the opportunities of young throwers to “overuse” their arms is critical in preventing pathologic conditions of the elbow.

Forces around the elbow during throwing. During the early and late cocking phases of throwing, significant distraction forces are placed on the medial elbow structures. This tensile force is transmitted to the medial UCL and medial epicondyle. In the pediatric population, the most vulnerable area of injury is the medial epicondylar physis, which can be avulsed with repetitive injury. In addition, the UCL can become stretched or incompetent, the flexor pronator mass can become strained, and the ulnar nerve can be stretched, resulting in neurologic symptoms. Whereas distraction forces are applied on the medial elbow, compressive forces are encountered on the lateral elbow structures during the early and late cocking phases.^{17–19} Sequelae of this repetitive compression of the radial head against the capitellum can result in growth disturbances, osteochondral fractures and loose bodies, and altered growth of the radial head.^{17,18}

During the acceleration phase of throwing, extreme pronation of the forearm results in a tension force on the lateral and posterolateral elbow structures. Lateral epicondylitis may result from repetitive overuse and strain of these muscles during this phase of throwing. During follow-through, hyperextension of the elbow places strain on the olecranon process and the anterior capsule. In late adolescence and into young adulthood, this stress can manifest as posteromedial elbow spurs, posterior/triceps spurs, and traction spurs of the coronoid process. Classic

overload symptoms of the elbow result from forces generated during the throwing motion and are helpful in determining pathologic findings: tensile overload of the medial restraints, compression overload of the lateral elbow, shear forces of the posteromedial elbow, and extension overload of the lateral restraints.^{18,26}

OVERVIEW OF PATHOLOGIES

Little Leaguer's Elbow

Little leaguer's elbow, or medial apophysitis, is a broad term used to describe a spectrum of pathologic conditions that can occur in the young baseball player or thrower. Although we typically associate this spectrum of injury with baseball, a wide range of throwing or overhead sports, including javelin, tennis, and football, can present in a similar fashion. The throwing motion in the developing athlete is associated with significant force across the medial elbow.¹⁸ During the throwing motion, and particularly in the late cocking phase of throwing, a valgus traction force is created on the medial elbow structures (i.e., the medial epicondyle, epicondylar apophysis, and medial ligament complex), whereas a compressive force is transmitted to the lateral side of the elbow (radial head and capitellum). With repetitive stress to the medial elbow, microtrauma and potentially degeneration of these structures may occur.²⁷ The subsequent tissue breakdown can cause the characteristic spectrum of disease: delayed or accelerated growth of the medial epicondyle, traction apophysitis and fragmentation of the medial epicondyle, medial epicondylitis, osteochondritis of the capitellum, hypertrophy of the ulna, osteochondral injury to the radial head, and olecranon apophysitis with delayed physal closure.

Aside from overuse and repetitive stress to the medial elbow, young pitchers are believed to have altered mechanics relative to older athletes. Younger pitchers may initiate trunk rotation earlier in the throwing motion, which causes hyperangulation at the shoulder and increased torque along the longitudinal axis of the humerus during late cocking, predisposing them to elbow and shoulder pathology.²⁷ Early trunk rotation occurs before the scapula and humerus are properly positioned in space. The resultant increase in horizontal abduction of the shoulder during stride results in excessive angulation of the shoulder throughout the cocking phase, which can predispose the athlete to anterior shoulder instability and an elevation in the horizontal adduction force applied to the humerus. When obtaining the history, a focus on the athlete's prior education regarding proper throwing mechanics, initiation of pitch counts with rest periods, and supervision for manifesting symptoms are all helpful in determining the prevention strategies that were in place for the athlete prior to the development of symptoms and can help guide treatment. Young athletes may not present in a typical fashion, and the symptoms may range from medial elbow pain and decreased throwing effectiveness to decreased throwing distance.

History

Obtaining a thorough history in pediatric and adolescent patients can often be challenging. The clinician must assess the overall

affect of the child, as well as the importance of the sport in his or her daily life. Some children may underreport their symptoms, so they may continue playing, whereas other children may overreport their symptoms in situations in which they feel pressured to compete by family or friends or pressured by the requirements of the daily routine. The age of the patient is important because it may provide a clue to the potential diagnosis. Patients in their early childhood and in whom the secondary ossification centers have yet to appear are more likely to report pain originating from the medial elbow as a result of repetitive injury to the ossification center and apophysis.¹⁸ Throwing forces may impede the characteristic appearance and growth of these ossification centers. In adolescence (which terminates with the fusion of the secondary centers of ossification with their respective long bones), muscle mass and force generation during throwing dramatically increase. In this age group, avulsion of the medial epicondyle can occur with resulting fracture displacement (Fig. 134.10). Incomplete avulsions are more common near the end of adolescence as fusion of the medial epicondyle occurs. With repetitive stress to the medial epicondyle, nonunions and delayed unions may occur and cause continued pain into young adulthood. When the patient enters young adulthood (after fusion of all the secondary ossification centers), injuries to the muscles and soft tissue attachments of the elbow are more common. Injury to the UCL and strain of the flexor/pronator mass are characteristically seen as overuse syndromes in this patient cohort.

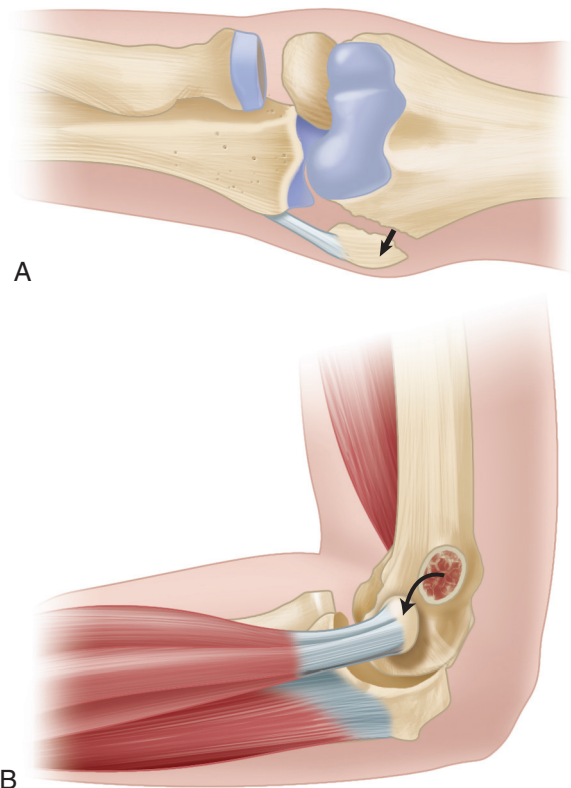


Fig. 134.10 (A and B) An avulsion fracture through the medial epicondyle, with attached ulnar collateral ligament and flexor/pronator mass.

In addition to age, the position of the throwing athlete should be elicited. Pitchers and quarterbacks place the most strain on their arms within their respective sports relative to other position players. In baseball, the likelihood of injury is more prevalent in pitchers and less likely in infielders, catchers, and outfielders, in decreasing order.¹⁷ Athletes who engage in throwing often injure themselves as a result of repetitive overuse rather than direct trauma, making hand dominance an important factor in determining injury.

In addition to this demographic information, the clinician should inquire about pain, which is often the presenting complaint. The onset, character, duration, intensity, location, persistence, and temporal association (e.g., night, day, or with activity) of the pain are important in both diagnosis and formulating a treatment plan. The onset of the pain can be gradual with an overuse injury pattern or more sudden if an avulsion fracture of the medial epicondyle is encountered. The pain may be dull or aching in patients with chronic injuries, or sharp and severe when a more identifiable, acute episode is the culprit.

The duration of the pain is important in determining prognosis. Short episodes of soreness are distinguished from pain that is present before, during, and after throwing, which is a more worrisome finding that could represent chronic overuse and portend a poorer prognosis. The location of the pain can help determine which structures are at risk. The temporal association of the pain is important in determining the most likely site of pathologic findings. Pain during the throwing motion in the late cocking phase typically implies medial overload or instability in a young adult or adolescent patient. Pain at rest or at night could indicate another underlying cause, and a neoplastic process should be ruled out in these instances.

In addition to pain, other symptoms may be present. Decreased ROM, swelling, decrease in velocity or control, and paresthesias or dysesthesias into the upper extremity, particularly the ulnar nerve, can also occur and should be evaluated. Decreased internal rotation of the shoulder has been associated with an increase in humeral retrotorsion seen in overhead throwing athletes, specifically those involved in baseball.²⁸ Increased humeral retrotorsion has been demonstrated in dominant throwing arms in youth and adult populations when directly compared with nondominant arms.^{28,29} In an adult pitching population, increased retrotorsion was shown to be a preventive adaptation for injury in the shoulder but a contributor for injury to the elbow.³⁰ Further studies regarding humeral retrotorsion and internal rotation deficits are needed to assess for the true effect on injury patterns in youth and adolescent throwers. After the symptoms have been elicited, a more detailed athletic history should also be obtained. The number of innings pitched, pitch count, type of pitches thrown, pitching rotation schedule, and training schedule can help determine if the athlete is placing his or her arm at risk for injury. Identifying a strain prior to ligament incompetence or growth plate disturbance is important to optimize treatment and prognosis. Associated injuries should also be evaluated, as well as any downstream (wrist/hand) or upstream (shoulder/neck) symptoms that could compromise successful treatment.

The surgical history of the elbow and shoulder should also be evaluated because altered biomechanics of both joints can place increased strain on the elbow during throwing. A thorough medical and family history should also be elicited for all patients with particular attention to any history of osteochondritis, Perthes disease, Kohler disease, and Osgood-Schlatter disease. When patients with any of these diseases participate in sports that increase the articular cartilage demand around the elbow, the likelihood of abnormal epiphyseal development is increased.¹⁷ In addition, a history of delayed skeletal maturation could cause a young child to throw on an age-determined team beyond his or her physiologic tolerance, which could lead to elbow abnormalities.

Physical Examination

The physical examination is first focused on the overall demeanor and affect of the young athlete. The examination should proceed in a fairly systematic fashion to decrease the risk of missing associated injuries. A complete examination of the cervical neck for range of motion (ROM), tenderness, and associated provocative tests (e.g., the Spurling test) can help determine cervical spine pathology. Both shoulders should be examined for strength, ROM, and stability. Abnormalities of the scapulothoracic joint, including scapular dyskinesia, should be sought, as should any evidence of internal rotation contracture of the glenohumeral joint, which can contribute to pathologic conditions of the elbow.^{20,30} Both elbows should first be inspected and evaluated for any asymmetry, deformity, or hypertrophy. Throwers often have hypertrophy of the muscles of their dominant arm or an alteration of the carrying angle. ROM of the elbows is assessed in flexion and extension, along with pronation and supination. The medial, lateral, and posterior elbow structures are palpated to elicit any tenderness. Because of the relative subcutaneous position of the osseous structures around the elbow, the medial epicondyle, lateral epicondyle, radial head, olecranon process, and collateral ligaments are readily identified and palpated to specifically identify any areas of discomfort. Patients with medial apophysitis typically demonstrate a flexion contracture greater than 15 degrees with point tenderness over the medial epicondyle.³¹

Palpation of the ulnar nerve is also performed in flexion and extension to evaluate for tenderness and subluxation. With slight flexion, the olecranon fossa can be examined, and gentle pressure can indicate pain over the posteromedial or posterolateral portion of the olecranon fossa. In slight flexion of 25 to 35 degrees, the olecranon is “unlocked” and the ligamentous stability of the elbow can be assessed. The lateral ligaments are assessed with slight varus and internal rotation stress applied to the arm, whereas the medial ligaments are assessed with valgus and external rotation stress applied to the arm.³² With the elbow flexed 90 degrees and the patient actively supinating, the distal biceps tendon can also be palpated in the antecubital fossa. The insertion of the triceps muscle is also easily identified on the olecranon process, and both of these muscles can be actively tested with flexion/supination and extension, respectively. Comparison is made to the contralateral extremity for any subtle differences that could indicate elbow instability or weakness. The examination

concludes with a detailed neurovascular examination of the distal extremity.

Imaging

Routine radiographs are an essential part of the evaluation of children and adolescents with elbow pain. Anteroposterior (AP), lateral, reverse axial, and comparison views are often obtained to rule out any osseous injury or irregularity of the growth centers. When injury to the ligaments is suspected, stress views can be obtained; however, a normal stress view does not rule out the presence of a significant ligamentous injury. Within the spectrum of little leaguer's elbow, a variety of pathologic conditions can be seen on radiographs. Fragmentation, enlargement, or fracture of the medial epicondyle can sometimes be appreciated.³³ Laterally, osteochondritis dissecans (OCD) can be appreciated as a lucency in the capitellum on oblique radiographs, or in advanced cases, degenerative arthritis and loose bodies may be visualized. Posteriorly, hypertrophy of the ulna may be present, which can impinge in the olecranon fossa of the distal humerus. With repetitive impingement, osteophytes and loose bodies may be visualized in the posterior compartment. Occasionally stress fractures of the ulna or olecranon apophysis can occur, in addition to delayed union at these sites. Classic findings of medial apophysitis include fragmentation and widening of the medial epiphyseal lines relative to the contralateral elbow.

Although most diagnoses can be confirmed with a thorough history, physical examination, and review of radiographs, other imaging modalities have been shown to increase sensitivity and the specificity of diagnosis. Ultrasound has been investigated as a potential imaging modality for evaluating OCD lesions of the capitellum and fragmentation of the medial epicondyle.³⁴ Bone scans can help determine areas of increased activity in persons with overuse injuries, and computed tomography (CT) scans can help better identify osseous anatomy in instances of fractures and loose bodies with osteophyte formation. Perhaps the most sensitive imaging modality is magnetic resonance imaging (MRI). MRI is helpful in evaluating the soft tissue structures of the elbow (UCL, biceps, triceps, and extensor/flexor tendon attachment sites), the articular cartilage (OCD, loose bodies, and avascular necrosis), and epiphyseal development (apophysis and epiphysis).^{35,36} In addition, in cases of OCD, MRI can help in decision-making with regard to surgical intervention. MRI often demonstrates more positive findings in patients with little leaguer's elbow than do traditional radiographs; however, these findings rarely alter clinical management of these patients.¹⁸

Decision-Making Principles

No hallmark sign accompanies the diagnosis of little leaguer's elbow; often it is the constellation of symptoms, physical signs, and confirmatory findings on radiographs or MRI that leads to the diagnosis. A high index of suspicion is warranted when any child presents with elbow pain and is involved in competitive sports that entail throwing. With prompt diagnosis and treatment, patients can often avoid more serious and long-term sequelae that are associated with this spectrum of injury. Perhaps equally as important is recognizing the variability

and normal findings that exist in this population. Hypertrophy of the throwing arm, valgus alignment of the extremity, and flexion contractures are fairly common in this patient population and by themselves rarely warrant treatment in asymptomatic patients.³⁷

Treatment Options

Treatment of little leaguer's elbow is tailored toward the specific diagnosis that is causing the pain or disability. Because this diagnosis can present along a continuum of injury, from a strain of the medial epicondyle to fracture and incompetence of the UCL, treatment is dependent on a multitude of factors: the injury severity, acuity, level of disability, and age. In patients who do not require acute surgical intervention and who have the capacity to heal, a period of rest is often warranted. In most cases, abstinence from throwing for 4 to 6 weeks results in the cessation of symptoms. The use of ice and nonsteroidal antiinflammatory medications has been shown to be beneficial in the acute stages. Currently no role exists for the use of injected corticosteroids in the management of medial apophysitis. In more severe cases, use of a removable posterior splint with immobilization may be warranted. After 6 weeks of rest, ROM is restored with gentle exercises and strengthening is begun. After 8 weeks, a throwing program is generally begun. Beyond rest, proper instruction on throwing mechanics may reduce the incidence of elbow injuries in these patients. In one investigation, it was found that youth who pitched with better pitching mechanics produced lower humeral internal rotation torque, lower elbow valgus load, and more efficiency during throwing.²⁴ This study has implications for the prevention of injury through better instruction. With regard to the frequency of pitching and the types of pitches thrown, a prior report demonstrated that the slider was associated with an 86% risk of elbow pain in youth who pitch, and a strong association was found between shoulder and elbow injuries and the number of games played and pitches thrown.³⁸ Development of the curve ball in youth baseball has been a controversial topic over the past few years. Initially it was thought to place significant stress across the medial elbow leaving athletes prone to injury.^{39,40} Several studies have since refuted this notion and have shown that the forces across the medial elbow are highest in the fastball.⁴¹⁻⁴³ Currently the US Baseball Medical & Safety Advisory Committee (USAB-MAC) advises against teaching the curveball before the age of 14, and the American Sports Medicine Institute (ASMI) identifies the curveball and slider as a potential injury risk in inexperienced pitchers.^{44,45} Perhaps the best treatment strategy known is prevention. Preventing excessive pitching in youth sports, enforcing rest days in between outings, discouraging year-round play, and watching for signs of fatigue will likely have a positive impact on the young pitcher's medial elbow. Fatigue has been shown to be a prominent risk factor for injury, and a recent study has shown that a loss of hip to shoulder separation is an early sign in adolescent pitchers.⁴⁶ Current guidelines have been developed for pitch counts in youth baseball as well, with the recommendation of pitches based on age. In 2010, Little League Baseball revised the previous guidelines set in 2007 that are in place today. Youths age 7 to 8 are allowed 50 pitches per day, which increases to 75 pitches per day at age 9 to 10. This

amount can be increased by intervals of 10 for every 2 years of chronologic age up to age 18. Pitchers 14 years and under are required to rest 1 day for up to 35 pitches thrown with a 1-day rest increase for every 15 pitches thrown thereafter. Ages 15 to 18 require 1 day of rest up to 45 pitches thrown with a 1-day rest increase for every 15 pitches thrown thereafter.^{47,48}

Results

Most patients with medial apophysitis return to sport at their preinjury level of competition with conservative treatment. No large-scale results of conservative management of this entity have been published. Because this diagnosis encompasses such a wide range of pathologic conditions, each distinct clinical entity has a varying prognosis with treatment.

Complications

With repetitive injury, after return to sport, the athlete is at risk for the development of growth disturbances around the elbow, persistent pain, avulsion fractures, OCD, and valgus instability after physal closure.

Osteochondritis Dissecans and Panter Disease (Osteochondrosis)

OCD is a focal lesion of the capitellum that typically occurs in adolescents between 13 and 16 years of age. Theories regarding the origination of this pathologic process vary, with some authors believing that the compressive forces from throwing result in valgus overload and contribute significantly to its development. Other investigations have pointed to the biomechanical mismatch between the capitellum and radial head as a causative mechanism. The etiology of this disease is multifactorial and likely results from microtrauma in the setting of cartilage mismatch and joint susceptibility.⁴⁹ Histology from patients who underwent surgery for OCD of the capitellum demonstrated damage to the articular cartilage surface as a result of repetitive stress after a degenerative and reparative process of articular and subchondral fracture.⁵⁰ Separation of the fragment occurs on the cartilage surface and may proceed to the subchondral bone in advanced stages.⁵⁰

Panner disease is a clinical entity that is distinct from OCD and is more common in younger patients. This clinical entity is a reversible, degenerative process of the capitellum that begins earlier in childhood, typically between the ages of 7 and 10 years, and involves the secondary ossification center of the capitellum. The hallmark of Panner disease is a process of degeneration followed by regeneration and recalcification.^{34,51} Potential causes of Panner disease have been attributed to endocrine disorders, fat embolism, congenital and hereditary factors, the tenuous blood supply of the capitellum, and repetitive trauma. In the child, the most common cause of lateral elbow pain is Panner disease; however, by adolescence OCD becomes the most common source of this pain. Panner disease has a rather acute onset, with fragmentation of the entire capitellar ossific nucleus.

History

Patients with OCD most commonly report elbow pain with decreased ability to throw and may localize their symptoms on

the lateral side of the elbow. The onset is typically insidious, and as the process progresses, loose bodies may form, which can cause the sensation of a “locking elbow.” Patients report joint stiffness and lack of ROM and may report catching within the joint. Patients with Panner disease may have difficulty describing their symptoms; however, it is usually an aching lateral-sided elbow pain that may be accompanied by swelling. Patients may report an inability to throw a ball or achieve terminal extension of the elbow. Most patients with either condition do not report a traumatic injury or other inciting event. It is important to identify causative mechanisms such as repetitive throwing, endocrine disorders, or family history that could implicate the diagnosis.

Physical Examination

In persons with Panner disease, the dominant arm is typically affected. On physical examination, patients may report tenderness over the lateral elbow and capitellum with slight effusion and synovial thickening. ROM is typically affected, with a lack of terminal extension of 20 to 30 degrees.⁵² Also noticeable may be a slight lack of pronation or supination with tenderness while attempting this movement.⁴⁹

In patients with OCD, the elbow may not demonstrate significant swelling. Because this clinical entity is commonly associated with little leaguer's elbow, many of the clinical manifestations are similar. Tenderness and laxity of the medial elbow, tenderness over the lateral epicondyle and capitellum, and clicking of the joint may be evident. In most patients, the pain may be difficult to localize because of the spectrum of pathologic conditions that is characteristic in these patients. Patients typically have diminished ROM, and in the setting of loose bodies, they may have acute, sharp pain with certain movements.

Imaging

In the setting of Panner disease, radiographs in the early stages of the disease demonstrate irregularity of the capitellum with areas of radiolucency and sclerosis, particularly near the physis. After several months, radiographs show enlarged areas of radiolucency with reconstitution of the bony epiphysis. After a period of 1 to 2 years, the epiphysis returns to a normal configuration without flattening. A recent meta-analysis showed no correlation between radiographic parameters and symptoms.⁵³ In some patients, the adjacent radial head may undergo early maturation compared with the contralateral elbow.

In the setting of OCD, supplemental radiographs such as oblique views or 45-degree flexed views can help better visualize the lesion in suspected cases. The classic finding in pediatric and adolescent OCD is a focal area in the anterolateral capitellum with rarefaction and irregularity of the articular surface. Sclerotic bone may surround the lesion, and loose bodies may be seen when the articular lesion becomes detached.³⁴ Healing of the lesion can take months to years and is visualized on radiographs as ossification of this radiolucent lesion. CT with or without arthrography may help better delineate the osseous anatomy and articular surface. MRI is the most sensitive test and may detect lesions prior to their appearance on radiographs. In lesions that appear early in the disease process, T1-weighted sequences

may detect decreased signal intensity within the lesion, with a normal appearance on T2-weighted sequences. Fat suppressed T2-weighted images are sensitive for detecting OCD lesions on the capitellum but maintain low specificity. Suspicious lesions detected on T2 fat suppressed sequencing can be further correlated on T1.⁵⁴ In more advanced cases, high signal intensity and cyst formation around the lesion could indicate impending detachment and loose body formation (Fig. 134.11).⁵⁵ More advanced MRI techniques with gadolinium may help define fragment stability and determine staging and prognosis.⁵⁶

Decision-Making Principles

In persons with Panner disease, the key to optimal management is proper diagnosis and recognition. When the clinician recognizes that Panner disease is usually a self-limited condition that disappears by adolescence, proper care and instruction can be provided to both the patient and family.⁵³ With OCD, several factors determine proper management and treatment. Classification systems of OCD have been devised to help standardize reporting and define treatment. In one of the first classification systems based on AP radiographs, the authors outlined three grades of OCD.⁵⁷ In grade I, radiographs demonstrated a translucent cystic shadow in the middle or lateral capitellum. In grade II, a clear zone or split line between the lesion and the adjacent subchondral bone is evident. In grade III, loose bodies are present.⁵⁷ This classification was shown to be the most reliable based on CT and radiography.⁵⁸ The International Cartilage Repair Society (ICRS) developed the following classification system for defining OCD lesions intraoperatively. ICRS OCD I indicates a stable lesion with a continuous but softened area covered by intact cartilage; ICRS OCD II, a lesion with partial discontinuity that is stable when probed; ICRS OCD III, a lesion with a complete discontinuity that is not yet dislocated; and ICRS OCD IV, an empty defect as well as a defect with a dislocated fragment or a loose fragment lying within the bed.⁵⁹ A



Fig. 134.11 A T2-weighted sagittal magnetic resonance imaging scan of a detached or semidetached osteochondritis dissecans lesion in a 12-year-old male adolescent.

recent MRI grading system utilizing T2-weighted imaging was proposed that accurately estimates the stability of an OCD lesion compared with intraoperative ICRS grading.⁵⁶ Treatment for these lesions is based on stability and is dependent on the ability of the lesion to heal based on the physiologic condition of the elbow (Table 134.1).⁶⁰ Based on this classification system, lesions deemed stable heal to completion with nonsurgical management, whereas unstable lesions demonstrate improved results with surgery.⁶⁰ As the characterization of capitellar lesions evolves, it will become increasingly important to use this information to predict prognosis and guide treatment. Management of OCD of the elbow is primarily determined by the integrity of the articular cartilage and the stability of the lesion. Unstable lesions are typically managed with surgical intervention, whereas stable lesions demonstrate more intrinsic ability for healing.

Treatment

In general, Panner disease is self-limited and treatment is largely symptomatic. Activity modification, rest, and antiinflammatories have been reported treatment options, with a majority of patients obtaining resolution of symptoms without morbidity.

In the early stages of OCD, treatment is largely nonoperative. Early-grade, stable lesions in patients with open physes are typically managed with a period of rest and cessation of sports participation. The amount of time that the patient refrains from participating in sports is determined by the length of time that symptoms persist. The typical duration is 3 to 6 weeks, followed by a 3- to 6-month period of progressive strengthening and ROM until full participation is achieved. Bracing and the use of nonsteroidal antiinflammatory medication may have a role; however, these treatments are largely supportive and have not been shown to provide tremendous benefit in the literature.^{60,61} In a small series, low-intensity pulsed ultrasound has been shown to shorten the repair period in nonoperatively treated patients.⁶²

Surgery for OCD of the elbow is typically reserved for patients who have closed physes, loose bodies, mechanical symptoms, unstable lesions on radiographs or MRI, or who have failed to respond to nonoperative management after 6 months and have a stable lesion. The best surgical procedure for patients with OCD of the elbow is a topic of considerable debate. Surgical options include open or arthroscopic fragment excision (with or without abrasion arthroplasty, drilling, or microfracture), fixation of the unstable or displaced lesion, bone grafting,

TABLE 134.1 Characteristics of Osteochondritis Dissecans Lesions Based on Stability

Classification	Capitellar Growth Plate	Range of Motion
Stable	Open	Normal
Unstable	Closed	Restricted

Data from Ruchelsman DE, Hall MP, Youm T. Osteochondritis of the capitellum: current concepts. *J Am Acad Orthop Surg*. 2010;18(9):557–567.

osteotomy, and osteochondral autograft transplantation (the OAT procedure). Our preference is fixation of these fragments when they are larger and when adequate purchase can be obtained with a 3.0-mm headed cannulated screw. Screw removal is typically necessary 3 months after fixation and healing. When smaller lesions are present, loose body removal and microfracture of the defect is a good surgical option.

Results

The spontaneous healing rates of patients with OCD are varied, and the results in the literature vary from excellent to poor. Mihara et al.⁴⁹ found that 25 of 30 early-stage lesions healed at final follow-up, whereas only 1 of 9 late-stage lesions had healed. A significant correlation of healing with open physes compared with closed physes was noted. In another study with longer follow-up, the authors found that more than 50% of the patients with stable lesions treated nonsurgically had mild discomfort at a mean follow-up of 13.6 years.⁶¹ In a different study, 50% of patients treated nonsurgically had persistent elbow symptoms with activities of daily living and also demonstrated radiographic evidence of osteoarthritis at 12.6-year follow-up.⁶³ These results demonstrate the variability in the outcomes of these patients; however, proper assessment of the stability of the lesion may be the reason for the discrepancy in the data.

For surgical treatment, the results may depend on the technique used, the stage of the lesion, and the length of follow-up of the patients. For open excision of the fragment and débridement, Bauer et al.⁶⁴ demonstrated poor results at long-term follow-up, with 40% of patients reporting recurrence of symptoms and loss of elbow extension. These patients all had advanced lesions at the time of surgery. Other studies have demonstrated more encouraging results, with nearly 50% of patients returning to athletics.⁶⁰ Short- and midterm results of patients who have undergone arthroscopic débridement and marrow stimulation techniques such as drilling or microfracture are encouraging.^{65–67} A recent meta-analysis in adolescents demonstrated an average return to sport of 87%.⁶⁸ In one report, the authors studied three elite gymnasts who underwent arthroscopic débridement and microfracture after failure of an initial trial of nonoperative management. They found full ROM and return to sport in all three patients at 1-year follow-up with hyaline-like cartilage on postoperative MRI.⁶⁹ Improvements in Disabilities of the Arm, Shoulder, and Hand scores and symptoms were shown in another study; however, many of the patients demonstrated decreased ability to participate in some sports because of their elbow.⁷⁰ Lewine et al.,⁷¹ in a more recent study, was able to demonstrate resolution of clinical or radiographical findings in 15 of 21 patients. Seven of these patients were unable to return to their primary sport. In a large retrospective review of fragment fixation versus excision, the authors compared 12 patients who underwent fixation with 55 patients who underwent excision of the unstable lesion. The authors found that fragment fixation performed better than removal, and they recommended bone grafting for higher grade lesions.⁶⁰ A recent retrospective review demonstrated healing in 20 of 26 elbows with younger patient age and smaller lesions demonstrating superior results.⁷² Several other studies show improved outcomes using fragment fixation

with an average return to sport of 68%.^{72–75} Osteotomy is a surgical option in the management of these patients, but it is rarely performed in the United States. In Japan, one study demonstrated good outcomes in baseball pitchers, with six of seven returning to sports, and all seven demonstrated remodeling of the capitulum at 6 months with a mean increase in ROM of 12 degrees.⁷⁶ OAT has gained much attention as a treatment option for OCD due to the growing body of evidence over the past decade. Indications include lesions that engage the radial head, larger lesions, higher grade lesions, and laterally based lesions on the capitellum. Recent literature has suggested that larger and more lateral based lesions often perform poorly without aggressive surgical treatment.^{77,78} The short-term results of the OAT procedure in the literature are promising, with more than 90% of patients in most series returning to their preinjury level of function without radiographic degenerative changes.^{79–81} A recent retrospective review of highly competitive adolescent athletes undergoing OAT for unstable capitellar OCD lesions demonstrated a return to play (RTP) of 100% as well as an increase in ROM and reported outcomes. Osteochondral fragment fixation was compared with the OAT procedure demonstrating similar results with regards to ROM, outcome scores, and RTP; however there was a 50% reoperation rate in the fragment fixation group. All patients in the reoperation group had large widespread lesions, leading the authors to conclude that OAT may provide better stability in those lesions.⁷³

Complications

Patients with Panner disease have relatively few complications. Most patients demonstrate the natural history of the condition with progression to healing. A few may experience worsening symptoms, loose body formation, and progression to an OCD clinical scenario. With OCD, most complications arise from incomplete healing of the lesion. Progression to loose body formation, osteoarthritis, limited ROM, and disability can occur, and athletes should be counseled on the possibility of not returning to their preinjury level of sport. Surgical complications can include infection, stiffness, hardware failure, neurologic injury (affecting the ulnar nerve), and failure to restore normal congruity to the elbow.

Medial Epicondyle Avulsion Fractures

Medial epicondyle fractures are common fractures in children and adolescents, accounting for 11% to 20% of all elbow fractures in this patient population.⁸² In many instances, this fracture pattern is associated with elbow dislocation; however, in the throwing athlete, this situation is rarely the case, and it is more commonly associated with repetitive overuse and traction on the medial epicondyle from the flexor/pronator mass.⁸³ Alternatively, a valgus force on the elbow while falling on an outstretched hand can also cause this injury pattern. These fractures most commonly occur in boys, with a peak incidence in age of 11 to 12 years.^{82–85}

History

Patients who sustain this injury likely report acute, sudden pain in their elbow localized over the medial epicondyle. Patients may

have reported a subluxation event or feeling a pop in their elbow that precipitated the swelling and discomfort. The history of pain may include sudden elbow pain in the setting of chronic elbow symptoms. A thorough and detailed athletic history is warranted to associate the pathologic condition with the throwing motion, and the constellation of findings seen in little leaguer's elbow should be sought.

Physical Examination

Physical examination typically demonstrates swelling in the area of the elbow, loss of a normal elbow contour (in cases of dislocation), and crepitus with ROM. Diminished ROM is present, and associated neurologic symptoms may occur because the ulnar nerve lies in close proximity to the posterior aspect of the medial epicondyle. In patients who require surgery, elbow stability can be assessed with the gravity-assisted valgus stress test (Fig. 134.12).⁸⁶

Imaging

The diagnosis is typically made with standard AP, lateral, and oblique radiographs of the elbow. Evaluating displacement on the AP radiograph may be the most accurate method of diagnosis; however, several studies have demonstrated the difficulty of and low interobserver agreement with the assessment of displacement.⁸⁷ Recently, a distal humerus axial view was proposed, which is obtained by centering the beam on the distal humerus, 15 to 25 degrees from the long axis of the humerus above the shoulder. In a cadaveric study, the authors were able to demonstrate increased accuracy and reliability of true displacement when compared with AP, lateral, and internal rotation radiographs.⁸⁸ CT can be used to more accurately assess displacement in patients with these fractures; one study demonstrated that in cases in which the fracture was thought to be minimally displaced on radiographs, significant displacement was encountered on CT.⁸⁹

Decision-Making Principles

The management of medial epicondyle fractures is controversial. Traditionally, this injury has been managed conservatively, and good results have been reported.^{90–93} The decision to treat the fracture surgically is typically reserved for open fractures and injuries in which fracture fragments are incarcerated within the joint space. Surgical fixation may be indicated for severely displaced fragments, cases in which valgus instability is present in

the athlete who places a high demand on the elbow, and when the ulnar nerve is compromised. With regard to fracture displacement as an indication for surgery, some authors advocate fixation for fragments with as little as 2 mm of displacement, whereas other authors recommend this treatment for fragments demonstrating greater than 15 mm of displacement.^{91–93} Currently there is no consensus on the amount of displacement required to necessitate surgical fixation, nor are there studies correlating the amount of displacement to clinical outcomes. Operative and nonoperative treatment in the setting of acute avulsion fractures among adolescent pitchers demonstrated equivalent outcomes and RTP.⁹⁴ Integrating the clinical history, examination, radiographs, and demands of the patient are all important in selecting the proper treatment for patients in this setting.

Treatment

Our preferred nonoperative treatment consists of immobilization in a long-arm cast with the elbow flexed to 90 degrees for 4 weeks. After removal of the cast, a splint can be applied with progressive ROM and return to sport after union is achieved. Surgical techniques for fractures meeting surgical criteria include suture repair, Kirschner wire fixation, screw fixation, and excision of the fragment with suturing of the soft tissue to the medial periosteum.^{95,96} Recent literature has described the use of resorbable devices and suture anchor fixation.^{97,98} Goals of surgery are to recreate the normal stability and mechanics of the elbow, and to create an epiphysiodesis of the medial epicondyle. Rigid internal fixation with screws (3.0, 3.5, or 4.0 mm, depending on the size of the epicondyle fragment) allows for earlier mobilization and may provide optimum stability.⁹⁰ Postoperatively, patients can be managed with a posterior splint, gentle exercises, and RTP once union is achieved and symptoms resolve.

Results

Reported results of nonoperative and operative management have generally been acceptable in the medical literature, even when fibrous union is achieved.^{90–93} Up to a 90% nonunion rate has been reported with nonoperative management; however, these are largely asymptomatic.⁹⁹ A shift toward operative treatment in recent years is likely due to the emerging concept of medial elbow stability in throwing and overhead athletes. If bony union is the primary outcome variable, surgical fixation provides better results in this regard, with a systematic review demonstrating 92% union in the surgically treated group and 50% union in the group managed without surgery.¹⁰⁰ Despite the higher union rates with screw or wire fixation, the patients managed without surgery achieved pain levels comparable with those of the surgical treatment group, with a trend toward less pain in the conservatively managed cohort. Functional outcome studies have demonstrated relative favorability for both treatment modalities, with some reports demonstrating improved functional results in patients managed without surgery. In a cohort of adolescent athletes including pitchers, 2-year follow-up data showed excellent outcome scores and a RTP rate of 100% in both the operative and nonoperatively treated groups.¹⁰¹ Although substantial valgus deformity occurs in fewer than 10% of patients, one study reported cubitus valgus as high as 35% after treatment.¹⁰² Some

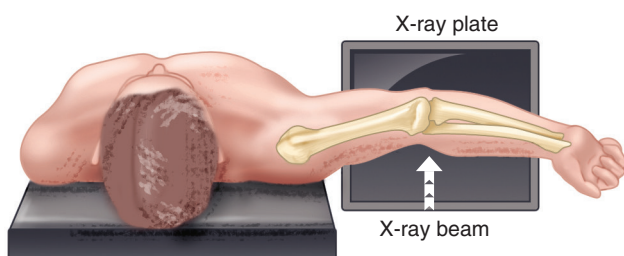


Fig. 134.12 The gravity stress test is shown. The arm is placed in full external rotation, permitting the weight of the forearm to deliver a valgus stress to the elbow.

studies have also demonstrated loss of ROM after both surgical and nonsurgical treatment (37 degrees and 15 degrees, respectively).⁸³ Many studies lack power due to insufficient sample size.

Complications

Complications from either treatment modality include loss of ROM, nonunion, and cubitus valgus. Two studies have obtained successful treatment of symptomatic nonunions, with one using open reduction internal fixation with cannulated screws and the other a tension band construct with bone grafting.^{103,104} Surgical complications include myositis ossificans, septic arthritis, pin tract or wound infections, and radial nerve injury.¹⁰⁵

Medial Collateral Ligament Injury

Injuries to the UCL, although uncommon in children and adolescents, have an increasing incidence as muscle mass and force increase in young adults. UCL injuries occur more commonly in adults but have been seen and treated in young adults and adolescents. Recently, the overall incidence of ulnar collateral reconstruction has risen, especially in the young athlete.^{106,107} Multifactorial reasons for this rise include increased participation in year-round athletics and lack of adherence to established guidelines. A recent survey of more than 750 young male pitchers found that despite the ASMI guidelines; 45% participated in a league with no pitch count or limits.⁴⁰ This spectrum of injury is closely associated with little leaguer's elbow and is part of the spectrum of injury that can be seen with valgus stress placed on the elbow during sports that entail throwing, particularly pitching.

History

Most patients have an insidious progression of discomfort and report pain over the medial elbow for months or even years before the ligament is definitively torn. Ruptures of the ligament occur as a sudden traumatic event, after which the elbow is painful enough to warrant cessation of throwing. Often patients report preexisting problems with elbow pain prior to this catastrophic event. Patients with overuse or attenuation of the ligament report medial elbow pain that is exacerbated by throwing, particularly in the late cocking and acceleration phases of throwing.¹⁰⁸

Physical Examination

On physical examination, patients demonstrate tenderness along the course of the UCL, from the medial epicondyle to the sublime tubercle. Subtle findings of instability are often present, demonstrated by flexing the elbow to 25 degrees to unlock the olecranon from its fossa and gently stressing the medial side of the elbow. This test assesses the competency of the anterior band of the UCL, as does the moving valgus stress test.¹⁰⁸ The posterior band of the ligament is tested with the milking maneuver, which is performed by pulling the patient's thumb with the forearm supinated, shoulder extended, and elbow flexed more than 90 degrees. Typically, pain with these maneuvers can be indicative of subtle instability and is deemed a positive finding. UCL injuries can commonly be associated with existing elbow or shoulder pathology including apophysitis and tendonitis, so a thorough evaluation of both joints is necessary.

Imaging

Plain radiographs are typically normal, except in cases of chronic injuries where the findings of valgus extension overload are found, such as posterior spurring and osteophyte formation. In patients with chronic valgus instability, excessive forces are placed on the posteromedial elbow during terminal elbow extension, which can result in degenerative changes of the posteromedial compartment, manifesting as spurring and joint space narrowing on routine radiographs. Stress views can be obtained, and relative widening of greater than 2 mm (compared with the contralateral extremity) can be considered a positive finding and indicative of instability. Comparison views may be necessary to more accurately determine pathology. A recent retrospective study found that patients diagnosed with a UCL injury had a significantly varus carrying angle and valgus distal humeral articular surface angle when compared with the control group.¹⁰⁹ Stress ultrasonography has been utilized to demonstrate progressive thickening of the UCL as years of professional pitching increase; however, these findings are limited in adolescent pitchers.^{110,111} MRI is the most sensitive test to evaluate the UCL and also helps delineate concomitant pathologic conditions in the area of the elbow. MRI and MRI arthrography may demonstrate thickening of the ligament in many asymptomatic patients. In patients with instability, the ligament may demonstrate attenuation or a tear from either the medial epicondyle or the sublime tubercle. Chondromalacia of the posteromedial compartment could indicate chronic valgus instability and may also be detected on MRI.^{112,113}

Decision-Making Principles

When considering surgical versus nonsurgical treatment, many factors go into determining the optimum strategy. Patients are likely surgical candidates when they have experienced an acute rupture, have signs of chronic instability, and have failed to respond to conservative treatment. Surgical treatment is best reserved for patients who wish to continue participating in sports that place a high demand on the medial elbow, including sports that entail throwing and gymnastics.^{108,114}

Treatment

In most instances, conservative treatment is warranted, which consists of rest, immobilization, use of nonsteroidal antiinflammatory medication, and physical therapy to restore strength, flexibility, and stability. The flexor and pronator muscle groups should be targeted because they are important secondary dynamic stabilizers to valgus stress around the elbow.¹¹⁵ Targeted evaluation of the throwing motion and biomechanics should be instituted by the coaching staff and trainers to identify any problems that could be contributing to the development of UCL abnormalities. In a mixed population of adult and adolescent athletes, treatment of partial tears of the UCL with autologous platelet-rich-plasma (PRP) injection followed by a targeted rehab protocol has been described with favorable results.^{116,117}

Surgical treatment for UCL insufficiency is offered after failure of a trial of 6 months of conservative management with persistent symptoms in the older athlete whose sport entails throwing. Complete tears of the ligament in young athletes typically require

surgical intervention. In select cases, direct surgical repair of the ligament can be attempted when direct avulsions off the medial epicondyle occur and the native ligament appears to be of good quality. In a study comprised of adolescent and college aged athletes, the authors showed a 97% return to sport rate with a failure rate of 6.7% with direct repair.¹¹⁸ Reconstruction of the ligament with a palmaris or gracilis tendon autograft is a better option when a tenuous ligament repair is imminent.¹¹⁹ Surgical reconstruction with a tendon graft is also indicated in the following circumstances: acute ruptures in throwers who lack enough remaining tissue ligament for a primary repair, the need to reestablish valgus stability in the presence of symptomatic chronic laxity, after débridement of calcific tendonitis in athletes if insufficient tissue is available to effect a primary repair, and when multiple episodes of recurring pain with throwing occur after periods of conservative care.^{120–122} Figure of eight tendon reconstruction, as advocated by Jobe et al.,¹¹⁴ or the docking technique, as advocated by Altcheck, are good options, and good results have been reported in the literature (Fig. 134.13).^{114,123,124}

Results

Many young patients respond favorably to a trial of conservative treatment. In one study of athletes that outlined the treatment of UCL insufficiency with conservative treatment, 42% of patients returned to sport;¹²⁵ however, no randomized prospective studies have been performed to compare this treatment method with surgical reconstruction. Early results of PRP injection for partial UCL tears show improved outcomes over conservative management alone; however, studies are limited.^{116,117} Prospective and retrospective studies evaluating the outcome of UCL reconstruction have demonstrated favorable results, with the majority of athletes returning to their preinjury level of participation. In one study evaluating the results in high school baseball pitchers, 74% of athletes were able to return to high-level throwing, which is comparable with the results seen in more mature athletes.¹²⁶ Jones et al.¹²⁷ performed UCL reconstruction utilizing the docking technique in a large group of adolescent athletes. Overall 87% of patients achieved an excellent result with inferior results noted in patients with concomitant OCD lesions. In a systematic review of UCL reconstructions, the authors found favorable results overall with success rates improved with use of a muscle-splitting approach, less handling of the ulnar nerve, and use of the docking technique. Lower complication rates were found, and increased RTP was found using the docking technique over the remaining

fixation techniques.^{128,129} Both the docking and double docking techniques have shown equivalent outcomes in recent studies, with no outcome differences in graft selection.¹³⁰ Overall, reconstruction results range from a 70% to 90% successful return to sport at the preinjury level.^{114,124,126,128,131}

Complications

Complications of surgical reconstruction include infection, fracture of the medial epicondyle and proximal ulna, ulnar nerve injury, and persistent pain and instability.

Posterior Elbow Pathologic Conditions (Posteromedial Impingement and Olecranon Osteochondrosis)

Injuries to the posterior compartment of the elbow are rare injuries in children and adolescents (Fig. 134.14). These injury patterns typically develop as late sequelae of chronic overuse syndromes (e.g., little leaguer's elbow and valgus extension overload). Pathologic conditions of the posterior elbow develop in response to chronic and repetitive extension overload from repetitive triceps contraction during the deceleration and follow-through phases of throwing. Posteromedial impingement is a term that denotes abutment of the medial aspect of the olecranon against the olecranon fossa as a result of valgus stress during terminal extension. Childhood injuries are more characteristically olecranon apophysitis and osteochondrosis with irregular ossification. In athletes approaching adolescence and young adulthood, these injuries are more commonly stress fractures or avulsion injuries of the olecranon apophysis with physeal widening, delayed fusion, or fragmentation.^{132,133} In young adults and after skeletal maturity, this spectrum of injury presents as posteromedial elbow impingement as a result of valgus extension overload, with osteophytes, loose bodies, and persistent elbow pain.¹⁰⁸ Recently it has been postulated that the result of repetitive posteromedial impingement in adolescents results in olecranon tip fractures as opposed to osteophyte formation seen in adult populations.¹³⁴

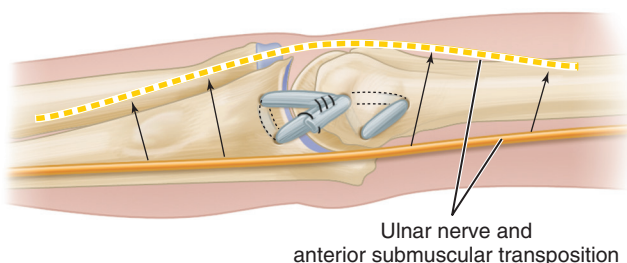


Fig. 134.13 An example of a medial collateral ligament reconstruction with anterior submuscular transposition of the ulnar nerve.

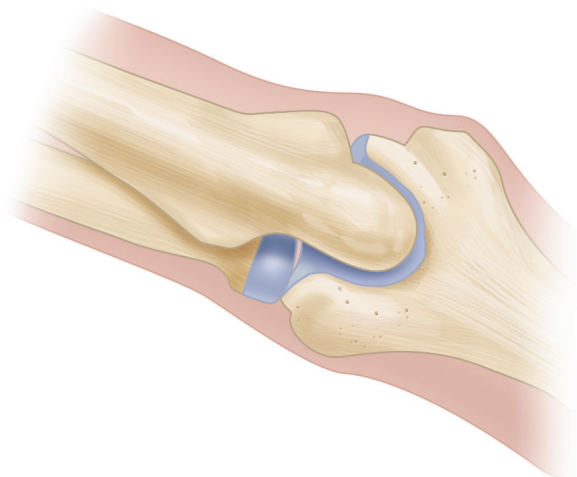


Fig. 134.14 The posterior compartment of the elbow.

History

Patients with pathologic conditions of the posterior elbow typically report pain in the posterior compartment of the elbow. This pain may be most noticeable, as the elbow is taken into extension or during the follow-through phase of throwing. Symptoms of locking or catching within the elbow joint could be a result of the presence of loose bodies.

Physical Examination

Physical examination shows pain in terminal extension, and patients may have tenderness to palpation over the posteromedial elbow and olecranon fossa. Because this injury can be the result of chronic valgus instability, assessment of the UCL should also be performed with stress testing as previously described. Tenderness over the proximal olecranon apophysis could indicate a stress fracture in this region, and distal triceps pain or pain with resisted elbow extension may indicate pathologic conditions within the extensor mechanism.

Imaging

Routine radiographs may demonstrate the presence of stress fractures or widening of the olecranon apophysis. These radiographs should be inspected for fragmentation, avulsions, or delayed fusion, which could indicate pathologic findings at this site. In cases in which injury is suspected, comparison views of the contralateral elbow can be obtained. With posteromedial impingement, osteophytes of the posteromedial olecranon may be visualized, along with loose bodies within the posterior compartment. Traction spurs of the proximal olecranon and triceps attachment are seen best on the lateral projection.^{135–137} MRI can be used to evaluate for concomitant pathologic conditions such as UCL tears, osteochondral lesions, and stress reactions, and can help locate loose bodies within the joint space.

Decision-Making Principles

Posterior elbow abnormalities are best managed based on the presenting diagnosis. The age of the patient often determines the most likely diagnosis in this setting. Younger patients are more likely to respond to conservative treatment, particularly when chronic insufficiency of the UCL has yet to develop. In older patients, avulsions of the proximal olecranon and triceps attachment and other injuries that denote chronicity may require surgical management.

Treatment

Treatment is tailored to the individual according to the age of the patient and the diagnosis. Osteochondrosis and stress fractures in younger patients usually respond to a period of rest (with or without immobilization), with progression to ROM and strengthening exercises after symptoms resolve. When returning to play, throwers are typically advanced with a rehabilitation program focusing on proper throwing mechanics and neuromuscular control, with progression to dynamic control and improvement of power. Avulsion¹³⁸ fractures with less than 2 mm of displacement can usually be treated conservatively with splint immobilization followed by functional ROM and strengthening

exercises. Small apophyseal fragments that do not compromise the extensor mechanism can be excised in patients who fail to respond to conservative treatment. Large avulsion fragments can be surgically treated with screw or suture fixation. In patients who have stress fracture nonunions or delayed union of the olecranon physis, open reduction and internal fixation with bone grafting is a viable option when a protracted period of conservative treatment fails.¹³⁵ In a cohort of adolescent overhead throwing athletes, two unique patterns of olecranon physeal nonunions were identified and treated successfully with internal fixation and autografting.¹³⁹ Removal of loose bodies and posteromedial osteophytes can also be performed for more mature patients who have persistent posteromedial elbow symptoms. A minimally invasive arthroscopic approach should be instituted in these cases to possibly optimize function and decrease morbidity. Posterior olecranon spur resection should be performed with care, because increasing bone resection creates more strain in the UCL with valgus load.

Results

The results of nonsurgical treatment are generally favorable, with most patients returning to sports after a trial of conservative treatment. Results of stress fracture and delayed physeal closure with surgical fixation have been reported in the literature in small case series with favorable outcomes in nearly all patients.^{135,139} Excision of spurs and loose bodies can provide symptomatic relief in most patients; however, addressing the underlying valgus instability through physical therapy or medial ligament reconstruction may be necessary to prevent further episodes. In a recent 2-year follow-up looking at arthroscopic resection of olecranon tip fractures, the authors found a good to excellent result in 85% of patients, with inferior results occurring in patients with undergoing concomitant UCL reconstruction.¹³⁴ The pediatric literature has a paucity of data regarding this treatment because most reports focus on adult athletes.

Complications

Complications can include nonunion, persistent pain, progression of posterior compartment osteoarthritis, and disability. Surgical complications include infection, wound complications, persistent nonunion, and progression of instability or iatrogenic instability as a result of surgical treatment.

Medial/Lateral Epicondylitis

A common anatomic location for pain in pediatric tennis players is the elbow, with 25% of 16- to 18-year-old boys and girls reporting previous or current elbow pain in survey data from the US Tennis Association.¹⁴⁰ Medial epicondylitis, lateral epicondylitis, or apophysitis of either epicondyles can be seen in skeletally immature and developing tennis athletes. Racquet sports place significant demand on the developing elbow as a result of repetitive eccentric contraction of the wrist extensors.¹⁴¹ This demand can also be seen in throwers during the follow-through phase of throwing.¹⁴² Repetitive microtrauma in patients with open physes can lead to apophysitis during childhood and the early adolescent period, with transition to epicondylitis in the late adolescent and young adult patient populations. Lateral epicondylitis has

been associated with equipment issues used in racquet sports, including grip size, metal racquets, tightly strung racquets, and racquets with increased vibration.^{143,144} Poor mechanics related to the backhand technique have also been associated with lateral epicondylitis. Proper¹⁴⁵ equipment sizing and instruction on proper mechanics of play are paramount in preventing this injury from developing and becoming a persistent problem for the young athlete. Anatomic studies examining the extensor carpi radialis brevis, which is implicated in lateral epicondylitis, have shown a unique anatomic location that makes its undersurface vulnerable to abrasion against the lateral edge of the capitellum during elbow motion.¹⁴⁶ Medial epicondylitis, although not as common as lateral epicondylitis, can be seen more commonly in baseball pitchers, golfers, and athletes who create significant valgus force at the elbow.¹⁴⁷ Valgus force created at the elbow can place increased strain on secondary stabilizers such as the flexor and pronator muscles, which arise from the medial epicondyle.¹⁴⁸ Poor technique with improper warm-up and fatigue can lead to injury and inflammation of the flexor/pronator mass. Implicated muscles of this muscle group are the flexor carpi radialis and pronator teres. In a prior study, the authors found macroscopic tearing of the flexor/pronator group in all of the patients who underwent surgical treatment for medial epicondylitis.¹⁴⁹ Just as with lateral epicondylitis, medial epicondylitis can be treated and hopefully prevented with appropriate equipment sizing, proper mechanics, warm-up, and stretching, so that symptoms do not persist into adulthood.

History

Patients with lateral epicondylitis typically report a burning-like pain on the lateral side of the elbow. Initially the pain is primarily activity related, and activities involving wrist extension may precipitate the pain. This injury is common in patients who play racquet sports, and this history should be elicited. Medial epicondylitis is similar in presentation to lateral epicondylitis, with symptoms arising from the medial elbow. Patients should be asked about exacerbating sports and activities for both conditions and if any associated neurologic symptoms are present, such as numbness and tingling, to rule out compression neuropathies.

Physical Examination

In persons with lateral epicondylitis, physical examination usually demonstrates pain and tenderness to palpation over the common extensor origin of the lateral epicondyle. Patients have pain with resisted wrist extension referred to the lateral side of the elbow. ROM is usually not affected. Persons with medial epicondylitis present with pain and tenderness to palpation over the medial elbow and may demonstrate increased pain with resisted wrist flexion. Pain may be associated with valgus stress testing of the elbow, which could indicate underlying valgus laxity as the cause of the pain.

Imaging

Routine radiographs are usually normal; however, in some cases, calcification may be present at the tendon origin. In younger patients, widening and fragmentation of the apophysis may be seen. MRI evaluation may prove useful in instances in which

there is a question of diagnosis. MRI may demonstrate increased signal on T2-weighted imaging or partial tearing of the tendon attachment of the lateral epicondyle. The reliability of MRI in detecting pathologic conditions was demonstrated to be excellent in a recent study, with good interobserver agreement; however, no correlation of MRI findings with symptoms was found in the patients studied.¹⁵⁰

Decision-Making Principles

Most young patients with medial and lateral epicondylitis are managed nonoperatively. Surgery is rarely indicated in this patient population. Most patients respond to conservative treatment measures and modification of activity with a short period of cessation of sports participation.

Treatment

Conservative treatment is mostly supportive for epicondylitis and apophysitis in the pediatric and adolescent patient populations. Cessation of sports participation and inciting activity is begun with or without a course of antiinflammatory medication and physical therapy. Therapy is aimed at restoring ROM, stretching the offending muscle group, and strengthening the muscles surrounding the elbow. Counterforce straps and wrist braces may offer additional relief from symptoms in patients who are not responding to other conservative measures. Additional therapies such as shock wave treatment, ultrasound, and light wave therapy are investigational and have shown variable results in the literature.^{151,152} Corticosteroid injections may be offered to young adults, although their utility in treating this condition has been questioned. A level I study in adults comparing autologous blood, corticosteroid therapy, and placebo saline solution injection for the treatment of lateral epicondylitis demonstrated no advantage of one treatment over the other.¹⁵³ A recent meta-analysis of randomized control trials showed no difference between corticosteroid injection and placebo injection.¹⁵⁴ The role of platelet-rich plasma is currently being intensely investigated, with some studies demonstrating improvement in patients with lateral epicondylitis.^{155,156} Surgery is rarely indicated in this patient population but can consist of débridement of the extensor or flexor tendon in refractory cases. Surgery can be performed with a percutaneous, arthroscopic, or open technique.

Results

Results of conservative treatment are generally favorable with both lateral and medial epicondylitis. In a systematic review evaluating the different types of therapy for lateral epicondylitis, the authors found that all types of therapy substantially improved the outcomes of patients, with eccentric strengthening being the most exhaustively studied.^{151,157} Eccentric strengthening exercises have demonstrated benefit in the literature; however, whether this treatment is better than stretching alone or stretching with concentric strengthening is unclear.^{152,158} A recent randomized study showed significant improvement with eccentric strengthening when compared with concentric strengthening with short-term follow-up.¹⁵⁹ Nonsteroidal antiinflammatory medications (both topical and oral) are routinely prescribed for persons with

epicondylitis; however, studies have shown both positive benefit and no benefit when compared with placebo.^{160,161} Current use of this modality is based on physician preference and the ability of the patient to tolerate treatment. Corticosteroid injections are still routinely used in practice for alleviating the pain of lateral epicondylitis. Although studies have shown acute benefit, most investigations that report long-term outcomes demonstrate no advantage to the use of corticosteroid for this condition.^{156,162–166} Surgical treatment, although rarely offered to young patients, has shown good results in the literature, regardless of the technique used, with one study showing improvement in outcomes in more than 90% of patients.^{149,167–172} Recently, arthroscopic techniques such as extensor carpi radialis brevis release have been studied and may provide small improved outcomes when compared to open techniques.¹⁷³ The treatment of medial epicondylitis mirrors the outcomes of lateral epicondylitis, with most patients responding favorably to nonsurgical management and good to excellent results in most patients who are ultimately candidates for surgical intervention.^{174,175}

Complications

Complications from conservative treatment are rare. Antiinflammatory medications may cause impaired renal function and gastrointestinal adverse effects, although in younger patients these effects are usually self-limiting. Corticosteroids may cause pigmentation changes of the skin and fat atrophy with a low risk of tendon rupture.¹⁷⁶ Three or greater corticosteroid injections prior to surgery have been shown to be most significant risk factor for revision surgery.¹⁷⁷ Surgical treatment complications include infection, iatrogenic elbow instability from excessive release, nerve injury, and heterotopic ossification.^{149,176,178,179}

SUMMARY

The treatment of elbow maladies in children and adolescents requires a thorough understanding of developing skeletal anatomy, biomechanics of sport, and a multitude of pathologic conditions that are characteristic in this patient population. With thorough history-gathering techniques, a comprehensive physical examination, and judicious use of musculoskeletal imaging, the diagnosis is usually readily delineated. Treatment is largely nonoperative in this patient population; however, understanding the need for surgery in select cases can help optimize the outcome.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

An KN, Morrey BF. Biomechanics of the elbow. In: Morrey, ed. *The Elbow and Its Disorders*. Philadelphia: WB Saunders; 1985.

Level of Evidence:

Book chapter

Summary:

This chapter focuses on the biomechanics of the elbow as it relates to anatomy, range of motion, and stability.

Citation:

Jobe FW, Stark H, Lombardo SL. Reconstruction of the ulnar collateral ligament in athletes. *J Bone Joint Surg Am*. 1986;68A:1158–1163.

Level of Evidence:

IV

Summary:

The authors of this case series outline the results of ulnar collateral ligament reconstruction in 16 athletes (most of whom were in sports entailing throwing). After surgery, 10 of the 16 athletes returned to their previous level of competition.

Citation:

Yadao MA, Field LD, Savoie FH III. Osteochondritis dissecans of the elbow. *Instr Course Lec*. 2004;53:599–606.

Level of Evidence:

Book chapter/instructional course lecture

Summary:

The authors of this comprehensive review outline the pathophysiology, diagnosis, and treatment of osteochondritis dissecans of the elbow joint. The article is a very good review of current concepts regarding this topic.

Citation:

Chen FS, Diaz V, Loebenberg M, et al. Shoulder and elbow injuries in the skeletally immature athlete. *J Am Acad Orthop Surg*. 2005;13:172–185.

Level of Evidence:

Review article

Summary:

The authors of this review article provide an overview of common shoulder and elbow maladies in pediatric and adolescent patients. Little leaguer's elbow, osteochondritis dissecans, epicondylitis, and other pathologic conditions are comprehensively reviewed.

Citation:

Nirschl RP, Pettrone FA. Tennis elbow: the surgical treatment of lateral epicondylitis. *J Bone Joint Surg Am*. 1979;61A:832–839.

Level of Evidence:

IV

Summary:

Tennis elbow surgery was performed for eight elbows in one of the first reports of this technique. An overall improvement rate of 97.7% was reported, with 85% of patients returning to rigorous activity.

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